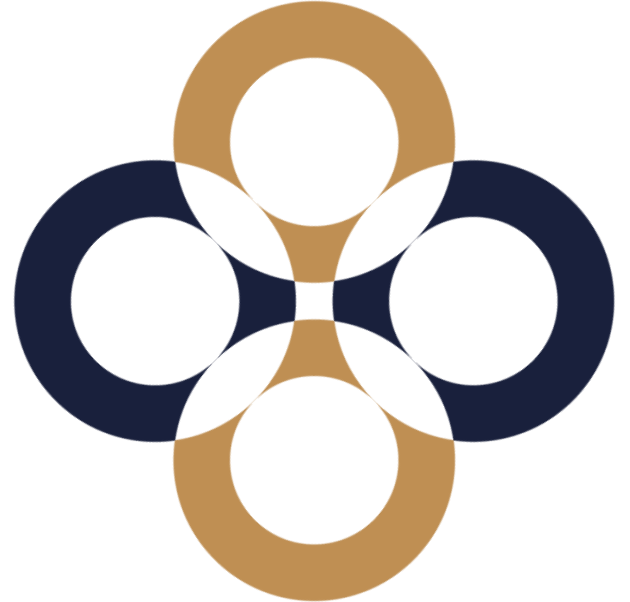


Business Cash Flow, Value drivers, Financial Planning

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Why do we need historical analysis?

Planning

Former performance maintainable?

Reviewing plans

Are assumptions realistic?

Performance measurement

Did we do better than required?



The three corrections of the statements

Main operation vs non-operative items

Rented office building

Standard vs Extraordinary items

Once in a lifetime grant

Operational vs Financing items

Management fees, exchange gain on interest

The balance sheet in a business view

Assets	Liabilities & Equity
Operational cash	Payables
Receivables	Short term operational liab.
Inventories	Financing debt (D)
Other operational current assets	
Invested assets needed for operation	Operational equity (E)
Non-operational assets	Non operational equity
	Non operational debt

Standard operation in the future

Long term outlook

Comparative
advantage
New entrants

Outstanding liabilities

Investments
Regulations
not yet
implemented

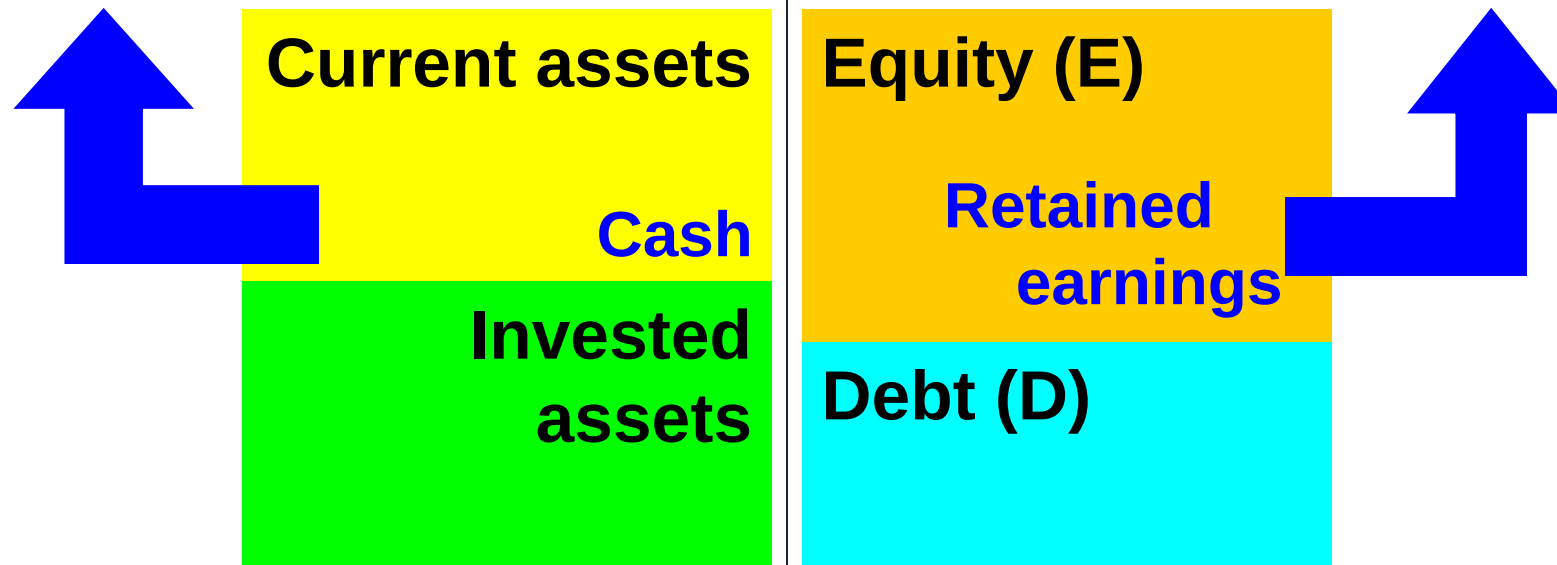
Ongoing issues

Litigations
Expected
changes in
regulations

Cash flow

Balance sheet

P/L statement



What is the last item in
the P/L statement?

ΔRetained earnings

P/L in a business view

Operational items	Non-operational items
Sales - Op. expenses (logic) = EBITDA - D&A = EBIT (<i>business view</i>)  + Non-operative profit (logic) + „Extraordinary” profit (logic) = Operational profit (accounting) Business cash flow	+ Profit of Financial Investments + Profit of Financing + Financial effects of operation = Profit from financial activities (accounting) = Gross profit - Corporate tax = Net profit - Dividend = Retained earnings

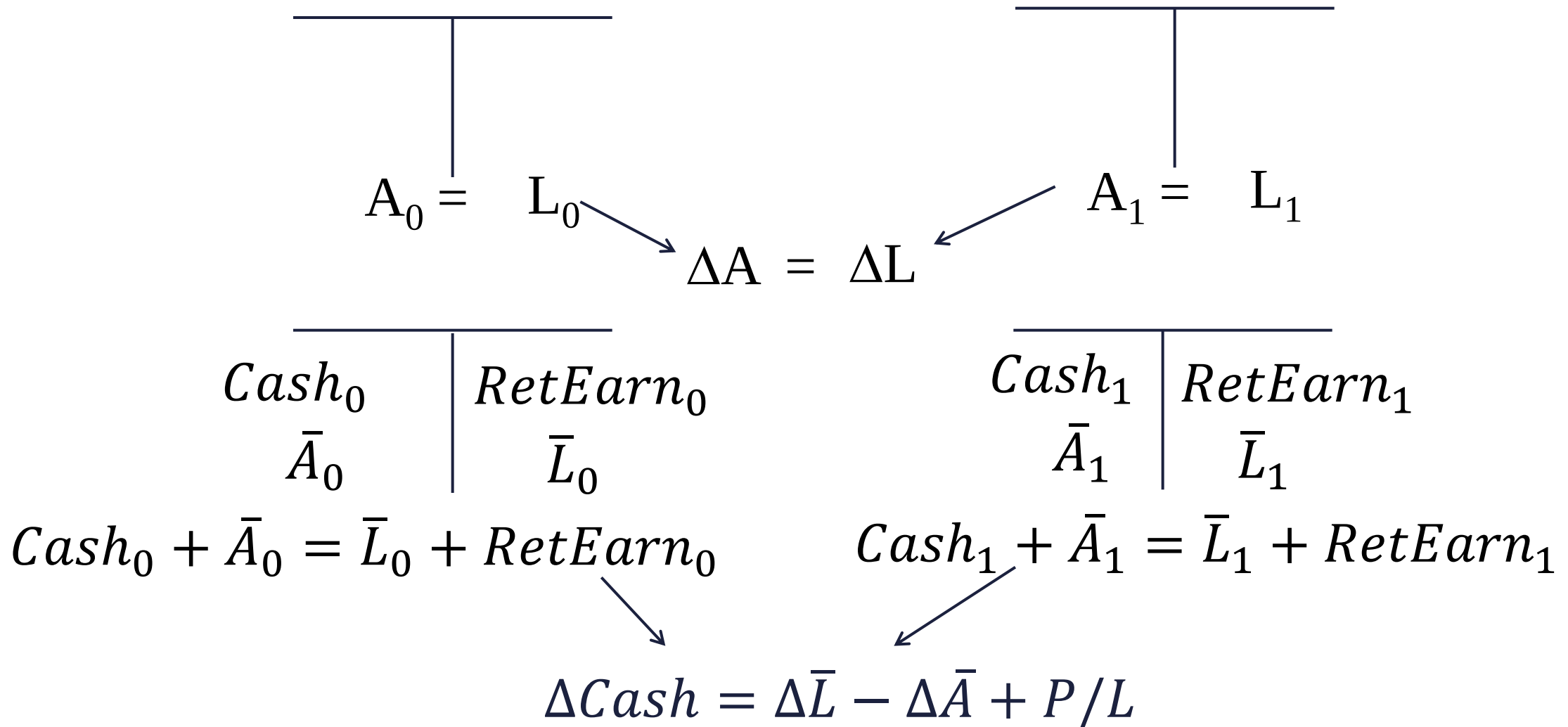
Direct

Directly tracing cash movements
Liquidity planning, short-term

Indirect

Builds on changes in Balance sheet items
Business planning, long-term

The logic behind the indirect cash flow



Operational items	Non-operational items
EBIT	+ Interest paid
- EBIT*t	- Tax shield
= NOPLAT	+ Change in Financing Debt
+ D&A	= FCFD
= Gross CF	FCFF – FCFD = FCFE
- Change in working capital	- Dividend payment
= Operating CF	+ (-) Share issue/buyback
+ Investment CF	= Change in Non-operating wealth (Not Change in Cash.)
= FCFF	

Difference between cash flows

	Accounting	Business
Application	Historical analysis, checking calculations	Predicting, supporting business decisions
Explained quantity	Change in cash	Change in non- operational wealth
Breakdown	Ordinary / Investment / Financial	Operational / Investment / Financing
Aim	Checking on change in cash	Quantifying sub-totals (FCFF, FCFE)

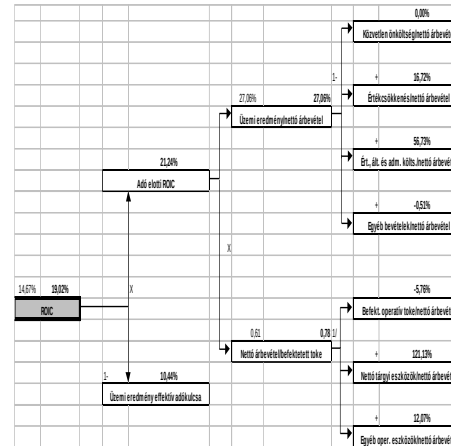
What are value drivers?

Operating parameters



Production, Logistics, Sales

Value drivers



Managerial Accounting, Controlling, Business planning

Intrinsic value



Corporate Finance

Share price



Investments

Quantities

1. ... linked directly to shareholder value creation,
2. ... representing both operation and financing,
3. ... covering the long term growth potential.

Value drivers 2

General use

Any quantity in direct connection with business performance (CF/employee).

Specific use

Quantities linked to the output a certain business area (customer satisfaction).

Valuation interpretation

Financial quantities used to describe the financial plan.

Key drivers of performance 1

$$V = \sum_{i=1}^{\infty} \frac{FCFF_i}{\prod_{j=1}^i (1 + WACC_j)}$$

$$g = \frac{NOPLAT_1}{NOPLAT_0} - 1$$

$$RONIC = \frac{\Delta NOPLAT}{\Delta IC}$$

$$ROIC = \frac{NOPLAT_1}{IC_0}$$

$$K = \frac{\Delta IC}{NOPLAT}$$

$$\Delta IC = IC_1 - IC_0$$

2021

Project 1	Equity
Project 2	Debt

Invested capital (**IC₀**)

2022

Project 1	Equity
Project 2	Debt
Project 3	

Invested capital (**IC₁**)

	2022	2023
EBIT	200	240
NOPLAT	100	120
Δ Working Capital	20	50
Δ Invested Assets	30	50
FCFF	50	20

Key drivers of performance 2

NOPLAT = Net Operating Profit Less Adjusted Taxes

g = annual growth rate of Noplat

IC = Invested Capital

ΔIC = annual change of IC, $\Delta IC_T = IC_{T+1} - IC_T$

Capital reinvested into the operation

FCFF = Free Cash Flow to Firm, available to distribute to financiers

$$FCFF_T = NOPLAT_T - \Delta IC_T$$

WACC = Weighted Average Cost of Capital,

artificial quantity to reflect average required rate of capital employed

Key drivers of performance 3

K = reinvestment rate

the theoretical proportion of NOPLAT reinvested into the operation:

$$K_T = \Delta IC_T / NOPLAT_T$$

ROIC = Return on Invested Capital :

$$ROIC_T = NOPLAT_T / IC_{T-1}$$

RONIC = Return on Newly Invested Capital,

(assuming the ROIC of the old project be unchanged):

$$RONIC_T = \Delta NOPLAT_T / DIC_{T-1}$$

Links among value drivers

$$\begin{aligned}\text{FCFF}_T &= \text{NOPLAT}_T - \Delta \text{IC}_T = \\ &= \text{NOPLAT}_T - \text{NOPLAT}_T * K_T = \\ &= \text{NOPLAT}_T * (1 - K_T)\end{aligned}$$

$$\begin{aligned}\text{RONIC}_{T+1} &= \Delta \text{NOPLAT}_{T+1} / \text{DIC}_T \\ K_T &= \text{DIC}_T / \text{NOPLAT}_T \\ g_{T+1} &= \text{DNOPLAT}_{T+1} / \text{NOPLAT}_T = \\ &= \text{RONIC}_{T+1} * K_T \\ K_T &= g_{T+1} / \text{RONIC}_{T+1}\end{aligned}$$

$$\text{FCFF}_T = \text{NOPLAT}_T * (1 - g_{T+1} / \text{RONIC}_{T+1})$$

$$\text{FCFF}_T = \text{IC}_{T-1} * \text{ROIC}_T * (1 - g_{T+1} / \text{RONIC}_{T+1})$$

A simple example

	2025
IC boy	1000
ROIC	
NOPLAT	
K	
FCFF	

Value drivers across the lifecycle

	Launch	Growth	Maturity	Decline
g_{Sales}	High	High	None	Negative
Operative profit margin	Low/High	High	Medium	Low
Corporate tax burden	Low	Low	Normal	Depends
Working capital need	High	High	Low	Decreasing
Invested asset need	High	High	Low/Medium	Low/Decreasing
r_A	Very high	High	Medium	Low
Planning horizon	Short	Medium/ Long	Long	Short/Medium
Proportion of terminal value	High	High	Medium	Medium

Business value by value drivers

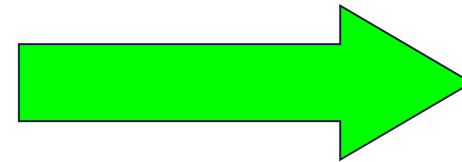
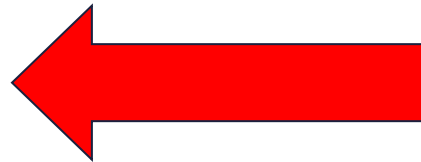
$$V_{OP} = \sum_{t=1}^{\infty} \frac{IC_{t-1} * ROIC_t * \left(1 - \frac{g_{t+1}}{RONIC_{t+1}}\right)}{\prod_{i=1}^t (1 + WACC_i)}$$

Assuming a growing perpetuity:

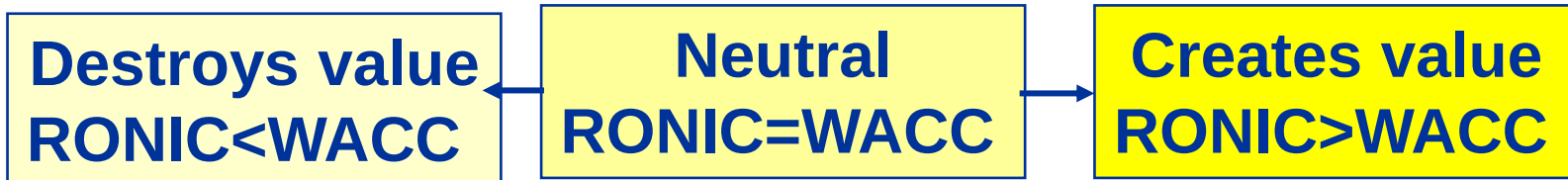
$$V_{OP} = \frac{NOPLAT * \left(1 - \frac{g}{RONIC}\right)}{(WACC - g)}$$

g, ROIC, WACC and the business value

		RONIC				
		6%	8%	10%	12%	14%
g	4%	709	891	1 000	1 073	1 125
	6%	495	811	1 000	1 126	1 217
	8%	215	706	1 000	1 196	1 336



Growth:



Measuring value added

Economic Value Added (EVA)

$$EVA = NOPLAT - WACC * IC = (ROIC - WACC) * IC$$

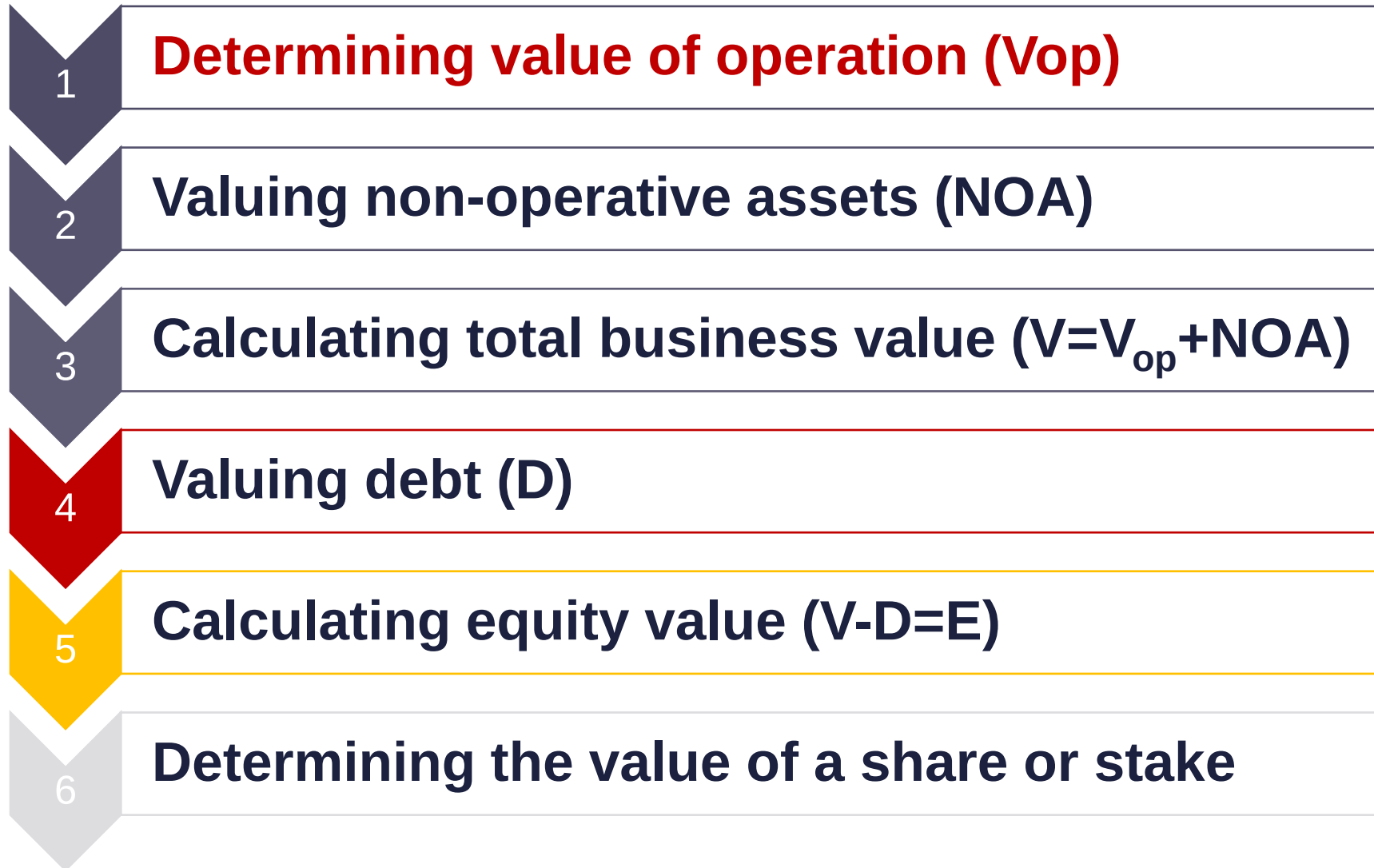
Residual Income (RI)

$$RI = \text{Net profit} - \text{Equity} * r_E$$

Market Value Added (MVA)

$$MVA = V - IC$$

The process of valuation



Determining value of operation 1

Business plan

1. Macro and industry analysis. Trends, outlooks, new entrants, regulations.
2. Competitors. Strategies, fields of business, organisation, financial performance, comparable ratios, competitive advantages, market power.
3. The business. Foundation, owners, mission, strategy, stakeholder policy, markets, operation, organisation.

Determining value of operation 2

4. Historical data. Firm data, financial reports, ratios.
5. Predictions. Future market conditions, growth in sales, cost structure, cash flow.
Consultation!
6. Valuation. What do we value? Choice of method, contrasting results, premiums & discounts.
7. Checking results. Assumptions used, market pricing vs result received, M&A data.

Which project / firm is better?

$$r = 10\%$$

$$\text{NPV}_A = 73.55$$

$$\text{NPV}_B = 54.13$$

$$\text{IRR}_A = 173.50\%$$

$$\text{IRR}_B = 208.26\%$$

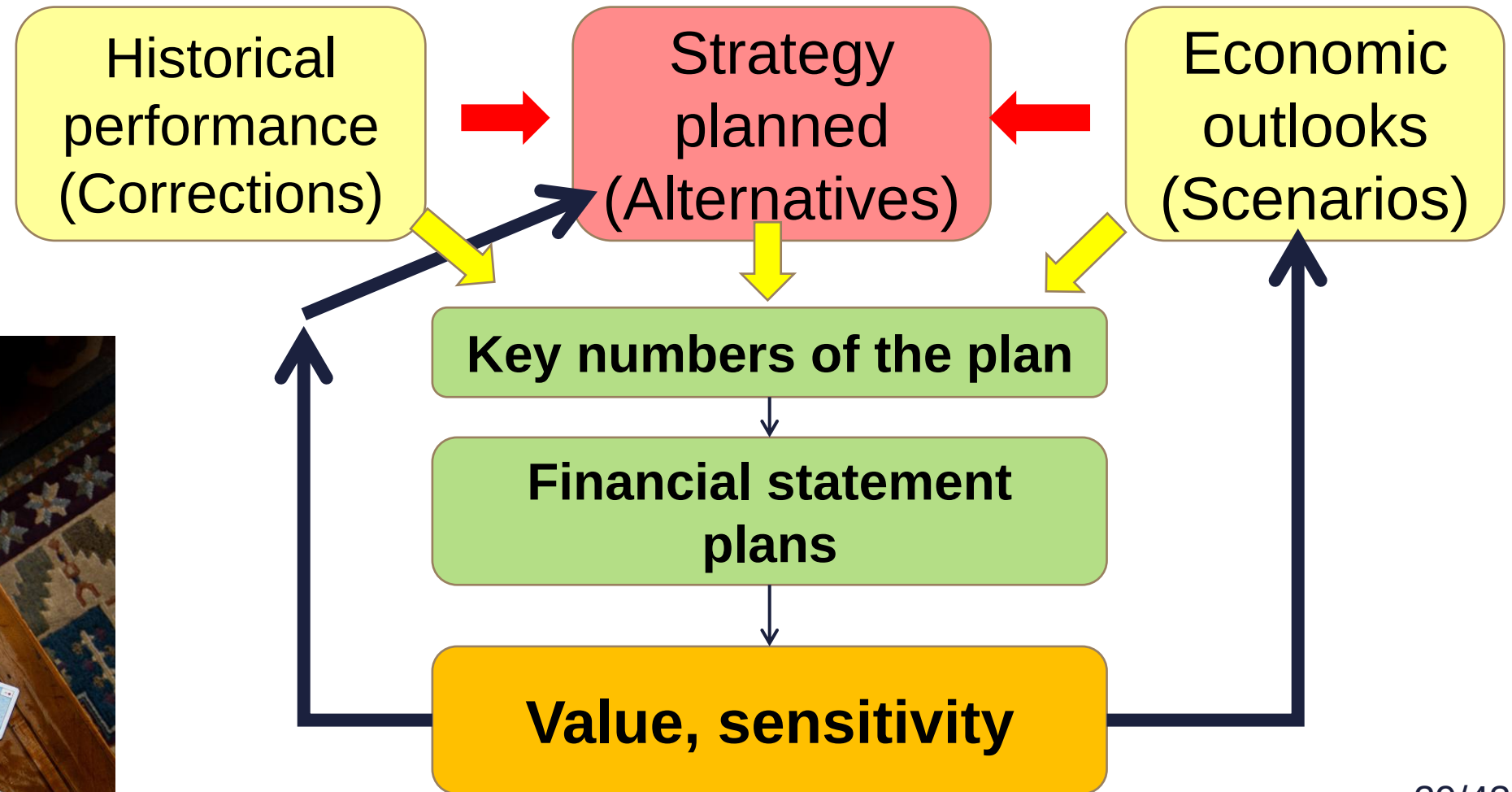
$$\text{PI}_A = 73.55\%$$

$$\text{PI}_B = 108.26\%$$

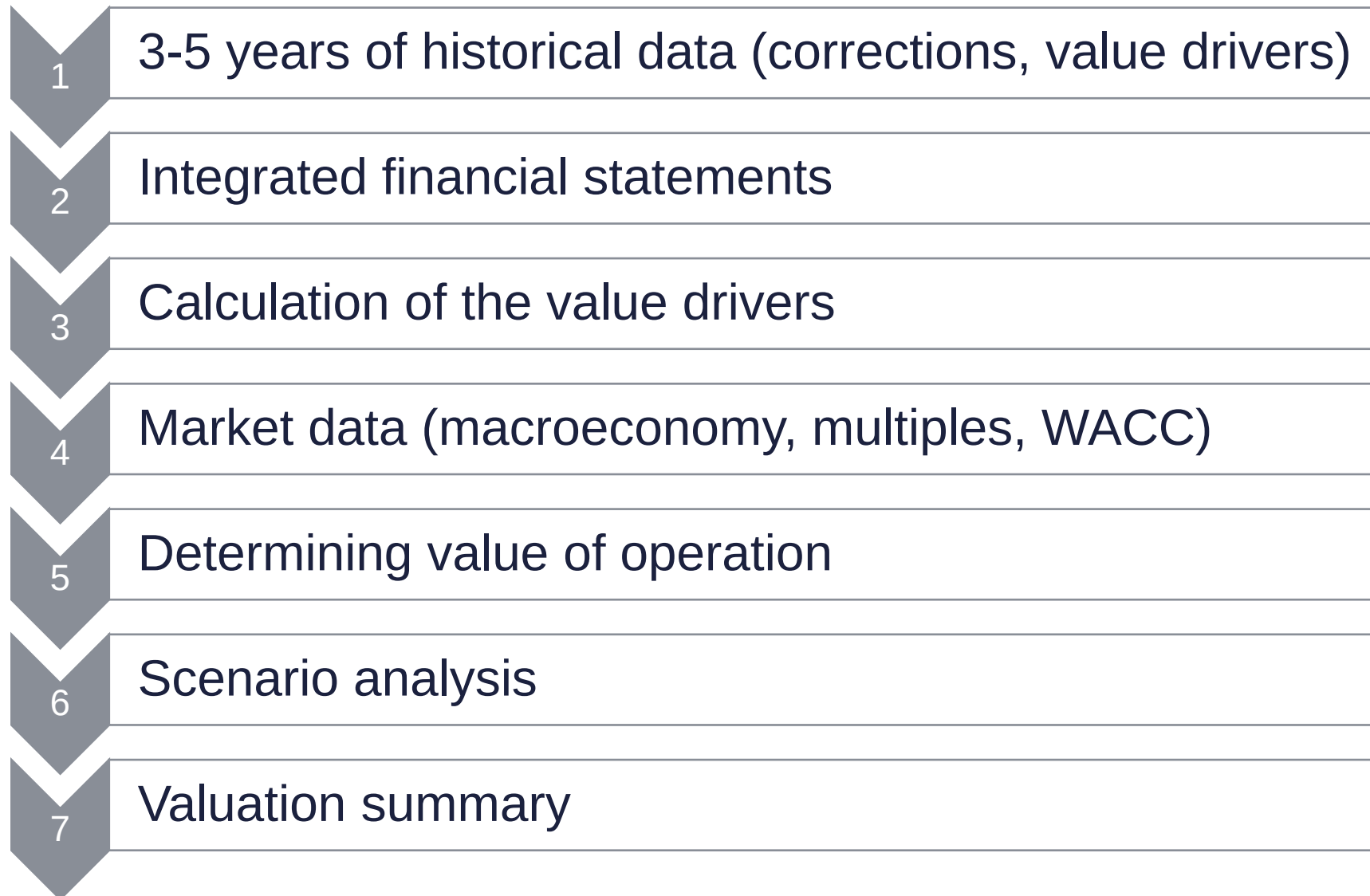
t	A	B
0	-100	-50
1	100	60
2	100	60

Better $\neq$ More valuable

The process of planning



Parts of the valuation model



Technical requirements

All inputs at one place

One directional logical connections

References instead of fix numbers

Key results are highlighted

Forecasting economic conditions

Porter five
forces

SWOT

PESTEL

Predicting
inflation

Yield curve

Market
outlooks

Development
of the
technology

Product life
cycle

Strategy

Steps of financial planning

1. Market performance

Quantity, price, conditions, timing $\Rightarrow$ Operational profit

2. Asset need

Invested and current assets, operational liabilities $\Rightarrow$ Total of Assets

3. Quantifying IC

Total of Assets - Operational sources $\Rightarrow$ IC need

4. Financing

Optimal sourcing of IC $\Rightarrow$ Total of Equity & Liabilities, finalising P/L

Predicting sales 1

Top-down approach: We derive the sales from the changes in the market conditions

logic: total market size, own share, development of prices

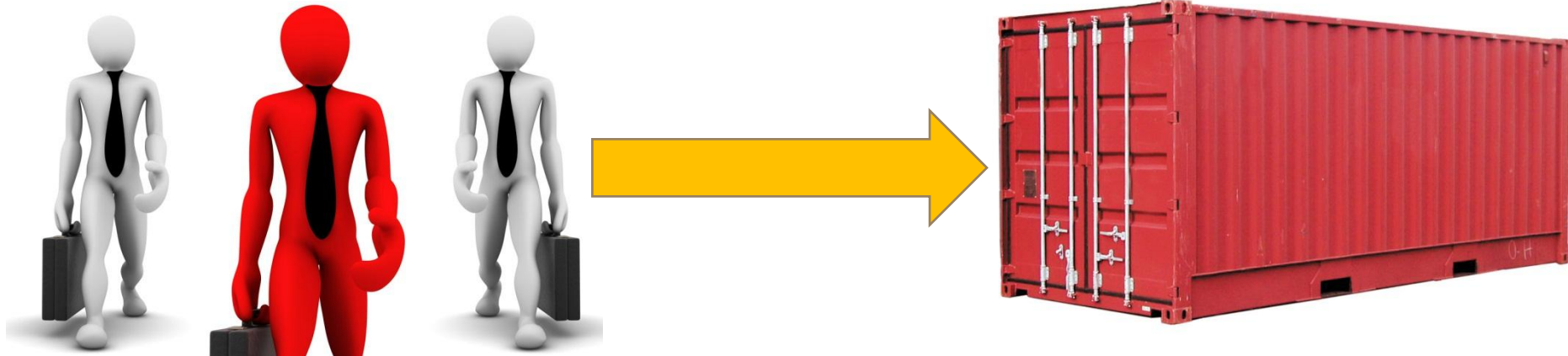
application: mature industries, big players, reliable market data available



Bottom-up approach: Prediction of demand for the current and potential buyers.

logic: demand data from customers, churn rates, acquisition possibilities

application: new industries, small players, application of data from comparable sectors



Predicting P/L

Type	P/L lines	Basis	Rate proposed
Operative	COGS	Sales (Q)	COGS / Sales
	SG&A	Sales (Q)	SG&A / Sales
	Material expenses	Sales (Q)	Material expenses/ Sales
	D&A	Invested assets at the end of the last year	D&A / Invested assets
Non- operative	Non-operative incomes	Linked non-operative assets	Non-op income / non-op assets
	Interest paid	Debt at the end of the last year	Interest paid / opening debt
	Interest received	Free cash at the end of the last year	Interest received / opening free cash

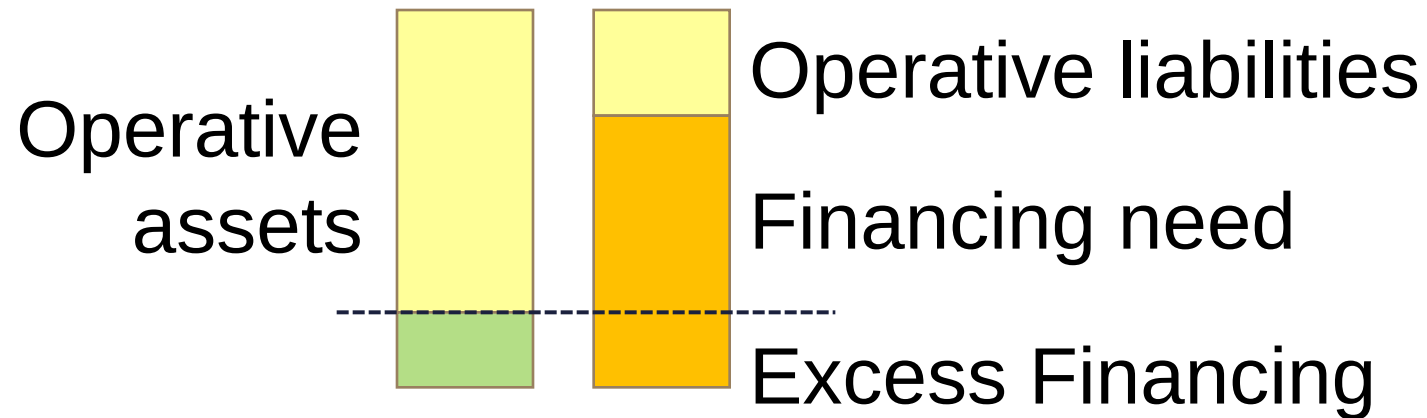
Predicting the balance sheet

Type	Balance sheet lines	Basis	Rate proposed
Operative	Operative cash	Sales	Operative cash / Sales
	Receivables	Sales	Average turnover days
	Inventory	COGS or Material + Labour expenses	Average turnover days
	Invested Assets	Sales (or quantity)	Invested assets / Sales
	Active accruals	Sales	Active accruals / Sales
	Payables	COGS or Material expenses	Average turnover days
Non-operative	Non-operative assets	None	Growth rate

Planning for financing

IC need is the difference of Assets and Operational liabilities:

$$\text{IC needed} = \text{Equity} + \text{Debt}$$



Assumptions on financing

Foreign capital

1. Remainder (balancing line),
2. Fix proportion: market average, percent of IC or V, own target value

Dividend

1. Remainder (based on reinvestment need),
2. Fixed proportion,
3. Nominal target (steady growth)

Which is the balancing line?

Debt

Operation,
Dividend, Equity
→ Debt (**Will we
get enough?**)

Cash

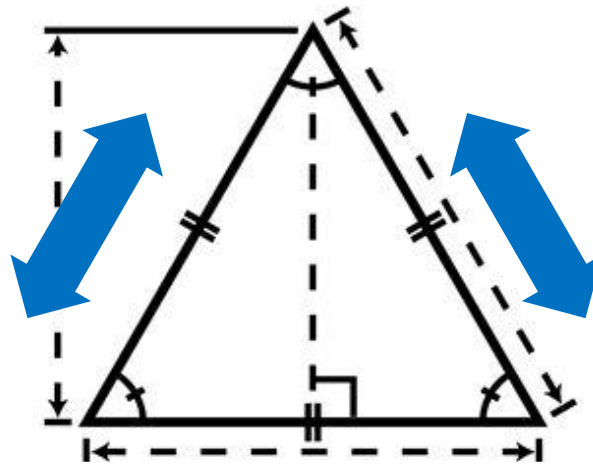
Operation,
Dividend, Debt
→ Cash (**Too
much?
Negative?**)

Dividend + Equity increase/decrease

Operation,
Financing,
Equity →
Dividend +
Capital
in/decrease

You can not set more than two factors

Operation and
investment policy



Dividend
policy



Financing
policy

Iteration for financing

Assuming start of the year debt adjustment may lead to circular referencing:

Interests paid/received → Net Profit → Dividend →
Reinvestment → Equity → Debt need/Extra cash →
Interests paid/received



**Thank you
for your attention.
Any questions?**

